

Basic propositional logic

Pablo Barenbaum

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In this note, we review the proofs of soundness and completeness for a Hilbert-style proof system for propositional logic. The proof of completeness relies on compactness.

1 Syntax

Definition 1 (Formulas). Let Var be a denumerable set of *propositional variables* $p_1, p_2, p_3, \dots$. The set Form of formulas is given by the grammar:

$$A ::= p \mid \top \mid \perp \mid \neg A \mid A \wedge A \mid A \vee A \mid A \rightarrow A$$

We write $\text{fv}(A)$ for the free variables of A . A context Γ is a set of formulas.

2 Semantics and compactness

Definition 2 (Valuations). A valuation is a function $V : \text{Var} \rightarrow \mathbf{2}$. Valuations are extended to formulas $-^V : \text{Var} \rightarrow \text{Form}$ as follows:

$$\begin{aligned} p^V &\stackrel{\text{def}}{=} V(p) \\ \top^V &\stackrel{\text{def}}{=} 1 \\ \perp^V &\stackrel{\text{def}}{=} 0 \\ (\neg A)^V &\stackrel{\text{def}}{=} 1 - A^V \\ (A \wedge B)^V &\stackrel{\text{def}}{=} A^V \cdot B^V \\ (A \vee B)^V &\stackrel{\text{def}}{=} A^V \sqcup B^V \\ (A \rightarrow B)^V &\stackrel{\text{def}}{=} (1 - A^V) \sqcup B^V \end{aligned}$$

where $\cdot$ denotes the product and $\sqcup$ denotes the maximum.

Definition 3 (Logical consequence). We write “ $\models$ ” for various notions of *logical consequence*:

- $V \models A$ holds if $A^V = 1$.
- $V \models \Gamma$ holds if $V \models A$ for every $A \in \Gamma$.
- $\models \Gamma$ holds if $V \models \Gamma$ for every valuation $V : \text{Form} \rightarrow \mathbf{2}$.
- $\Gamma \models A$ holds if $V \models \Gamma$ implies $V \models A$ for every valuation $V : \text{Form} \rightarrow \mathbf{2}$.

We say that Γ is *valid* or a *tautology* if $V \models \Gamma$ for every valuation $V : \text{Form} \rightarrow \mathbf{2}$. We say that Γ is *satisfiable* if $V \models \Gamma$ for some valuation $V : \text{Form} \rightarrow \mathbf{2}$, and *unsatisfiable* otherwise. Two formulas A, B are *logically equivalent* if for every valuation $V : \text{Form} \rightarrow \mathbf{2}$ $V \models A$ holds if and only if $V \models B$ holds. Logical equivalence is defined similarly for contexts.

Remark 4. $\Gamma \models A$ holds if and only if $\Gamma, \neg A$ is unsatisfiable.

Theorem 5 (Compactness). *The following are equivalent:*

1. Γ is satisfiable.
2. Γ' is satisfiable for every finite subset $\Gamma' \subseteq \Gamma$

Proof. (1 $\implies$ 2) Immediate. (2 $\implies$ 1) For each $n \in \mathbb{N}$, let Γ_n be the set of formulas in Γ that use at most the first n propositional variables, i.e.

$$\Gamma_n \stackrel{\text{def}}{=} \{A \in \Gamma \mid \text{fv}(A) \subseteq \{p_1, \dots, p_n\}\}$$

The set Γ_n may be infinite, but up to logical equivalence it contains at most 2^{2^n} formulas. So for each n there is a finite set $\Gamma'_n \subseteq \Gamma_n$ logically equivalent to Γ_n . Since Γ'_n is satisfiable, there is a valuation V_n such that $V_n \models \Gamma_n$ for all $n \in \mathbb{N}$. Note moreover that if $i \geq n$ then $V_i \models \Gamma_n$.

Now for each $n \in \mathbb{N}_0$ we inductively define a set $I_n \subseteq \mathbb{N}$ in such a way that

- I_n is infinite;
- if $i, j \in I_n$ then V_i and V_j coincide on the first n propositional variables; and
- $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$

To do so, set $I_0 \stackrel{\text{def}}{=} \mathbb{N}$ and

$$I_{n+1} \stackrel{\text{def}}{=} \begin{cases} E_n^1 & \text{if } E_n^1 \text{ is infinite} \\ E_n^0 & \text{otherwise} \end{cases}$$

where $E_n^b = \{i \in I_n \mid i \geq n+1, V_i(p_{n+1}) = b\}$.

Now we define a valuation $V : \text{Var} \rightarrow \mathbf{2}$ as follows:

$$V(p_n) \stackrel{\text{def}}{=} V_i(p_n) \quad \text{for any } i \in I_n$$

noting that if $i \geq n$ then $V_i(p_n) = V(p_n)$. To conclude, we claim that $V \models \Gamma$. Indeed, let $A \in \Gamma$. Then $A \in \Gamma_n$ for some $n \in \mathbb{N}$. In particular, take $i \in I_n$. Then $V_i \models \Gamma_n$ because $i \geq n$, so also $V_i \models A$. Moreover $A^{V_i} = A^V$ because A uses only the first n variables, so $V \models A$ as required. $\square$

Corollary 6. *If $\Gamma \models A$ then there is a finite subset $\Gamma' \subseteq \Gamma$ such that $\Gamma' \models A$.*

Proof. By the previous remark, $\Gamma \models A$ holds if and only if the set $\Gamma, \neg A$ is unsatisfiable. So if $\Gamma \models A$ by Compactness there exists a finite unsatisfiable subset $\Gamma_0 \subseteq \Gamma, \neg A$. Without loss of generality we may assume that $\neg A \in \Gamma_0$ so $\Gamma_0 = (\Gamma', \neg A)$ is unsatisfiable. This in turn means that $\Gamma' \models A$. $\square$

3 Hilbert-style proof system

Definition 7 (Propositional proof system). The only deduction rule is *modus ponens*:

$$\frac{\vdash P : A \rightarrow B \quad \vdash Q : A}{\vdash (P \cdot Q) : B}$$

Plus the following axioms, instantiated on arbitrary formulas $A, B, \dots$:

Implication

$$\begin{aligned} K & : A \rightarrow (B \rightarrow A) \\ S & : (A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow B) \rightarrow (A \rightarrow C) \end{aligned}$$

Truth and falsity

$$\begin{aligned} \text{trivial} & : \top \\ \text{abort} & : \perp \rightarrow A \end{aligned}$$

Negation

$$\begin{aligned} \text{negi} & : (A \rightarrow B) \rightarrow ((A \rightarrow \neg B) \rightarrow \neg A) \\ \text{nege} & : A \rightarrow (\neg A \rightarrow B) & \text{NB. Redundant if dneg is allowed.} \\ \text{dneg} & : \neg \neg A \rightarrow A & \text{NB. Classical.} \end{aligned}$$

Conjunction

$$\begin{aligned} \text{pair} & : A \rightarrow (B \rightarrow (A \wedge B)) \\ \pi_1 & : (A \wedge B) \rightarrow A \\ \pi_2 & : (A \wedge B) \rightarrow B \end{aligned}$$

Disjunction

$$\begin{aligned} \text{in}_1 & : A \rightarrow (A \vee B) \\ \text{in}_2 & : B \rightarrow (A \vee B) \\ \text{match} & : (A \rightarrow C) \rightarrow ((B \rightarrow C) \rightarrow ((A \vee B) \rightarrow C)) \end{aligned}$$

We write $\Gamma \vdash A$ if A can be proved using *modus ponens*, the axioms, and hypotheses from Γ . We assume that $\rightarrow$ is right-associative and $\cdot$ is left-associative. Application $P \cdot Q$ is written PQ .

Lemma 8 (Identity). $\vdash A \rightarrow A$ holds for any formula A .

Proof. It suffices to take $\text{id} \stackrel{\text{def}}{=} SKK$:

$$\frac{\frac{\frac{\vdash S : (A \rightarrow (A \rightarrow A) \rightarrow A) \rightarrow (A \rightarrow A \rightarrow A) \rightarrow A \rightarrow A \quad \vdash K : A \rightarrow (A \rightarrow A) \rightarrow A}{\vdash SK : (A \rightarrow A \rightarrow A) \rightarrow A \rightarrow A} \quad \vdash K : A \rightarrow A \rightarrow A}{\vdash SKK : A \rightarrow A}}$$

□

Lemma 9 (Weakening). *If $\Gamma \vdash A$ and $\Gamma \subseteq \Gamma'$ then $\Gamma' \vdash A$.*

Proof. Straightforward by induction on the derivation of $\Gamma \vdash A$. □

Theorem 10 (Deduction Theorem). $\Gamma, A \vdash B$ *if and only if* $\Gamma \vdash A \rightarrow B$.

Proof. The $(\Leftarrow)$ direction is immediate since by Weakening (Lem. 9) we have that $\Gamma, A \vdash A \rightarrow B$ so:

$$\frac{\Gamma, A \vdash A \rightarrow B \quad \Gamma, A \vdash A}{\Gamma, A \vdash B}$$

For the $(\Rightarrow)$ direction we proceed by induction on the derivation of $\Gamma, A \vdash B$. There are three cases, either (1) we use an axiom or an assumption from Γ , (2) we use A , or (3) we use the *modus ponens* rule:

1. **Axiom or fixed assumption.** That is, the derivation of $\Gamma, A \vdash B$ is of the form:

$$\frac{}{\Gamma, A \vdash B}$$

where B is one of the axioms (K , S , trivial, etc.) or $B \in \Gamma$. Then B can also be proved under Γ so we have:

$$\frac{\Gamma \vdash K : B \rightarrow A \rightarrow B \quad \Gamma \vdash B}{\Gamma \vdash A \rightarrow B}$$

2. **Identity.** That is, the derivation of $\Gamma, A \vdash B$ is of the form:

$$\frac{}{\Gamma, A \vdash A}$$

Then by the Identity lemma (Lem. 8) we have:

$$\frac{}{\Gamma \vdash A \rightarrow A}$$

3. **Modus ponens.** That is, the derivation of $\Gamma, A \vdash B$ is of the form:

$$\frac{\Gamma, A \vdash C \rightarrow B \quad \Gamma, A \vdash C}{\Gamma, A \vdash B}$$

So by *i.h.* on each of the premises we have:

$$\frac{\frac{\frac{}{\Gamma \vdash S : (A \rightarrow (C \rightarrow B)) \rightarrow (A \rightarrow C) \rightarrow (A \rightarrow B)}}{\Gamma \vdash (A \rightarrow C) \rightarrow (A \rightarrow B)} \quad \frac{i.h.}{\Gamma \vdash A \rightarrow (C \rightarrow B)}}{\Gamma \vdash A \rightarrow C} \quad \frac{i.h.}{\Gamma \vdash A \rightarrow C}$$

$$\frac{}{\Gamma \vdash A \rightarrow B}$$

□

3.1 Basic facts

Following, we state and prove some basic principles.

1. **Contrapositive** ($\Gamma, A \rightarrow B \vdash \neg B \rightarrow \neg A$).

By the Deduction Theorem (Thm. 10), it suffices to show that $\Gamma, x : A \rightarrow B, y : \neg B \vdash \neg A$. Indeed:

$$\Gamma, x : A \rightarrow B, y : \neg B \vdash \text{negi } x \left(\underbrace{K y}_{A \rightarrow \neg B} \right) : \neg A$$

2. **Explosion principle** ($\Gamma, A, \neg A \vdash B$). This is an immediate consequence of the Deduction Theorem (Thm. 10) and axiom **nege**. Moreover, if **dneg** is allowed, **nege** is redundant:

$$\Gamma, x : A, y : \neg A \vdash \text{dneg} \left(\underbrace{\text{negi} \left(\underbrace{K x}_{\neg B \rightarrow A} \right) \left(\underbrace{K y}_{\neg B \rightarrow \neg A} \right)}_{\neg \neg B} \right) : B$$

3. **False antecedent** ($\Gamma, \neg A \vdash A \rightarrow B$). Immediate from the explosion principle and the Deduction Theorem (Thm. 10).
4. **True consequent** ($\Gamma, B \vdash A \rightarrow B$). Immediate from the axiom $K : B \rightarrow A \rightarrow B$ and the Deduction Theorem (Thm. 10).
5. **Disproving an implication** ($\Gamma, A, \neg B \vdash \neg(A \rightarrow B)$).

By the true consequent property we have that $\Gamma, A, \neg B, A \rightarrow B \vdash B$ so by the Deduction Theorem (Thm. 10) $\Gamma, A, \neg B \vdash P : (A \rightarrow B) \rightarrow B$. Then:

$$\Gamma, x : A, y : \neg B \vdash \text{negi} \left(\underbrace{P}_{(A \rightarrow B) \rightarrow B} \left(\underbrace{K y}_{(A \rightarrow B) \rightarrow \neg B} \right) \right) : \neg(A \rightarrow B)$$

6. **Conjunction introduction** ($\Gamma, A, B \vdash A \wedge B$). By the Deduction Theorem (Thm. 10) and the axiom pair.
7. **Negated conjunction introduction** ($\Gamma, \neg A \vdash \neg(A \wedge B)$ and $\Gamma, \neg B \vdash \neg(A \wedge B)$).

$$\Gamma, x : \neg A \vdash \text{negi} \left(\underbrace{\pi_1}_{(A \wedge B) \rightarrow A} \left(\underbrace{K x}_{(A \wedge B) \rightarrow \neg A} \right) \right) : \neg(A \wedge B)$$

Similarly:

$$\Gamma, y : \neg B \vdash \text{negi} \left(\underbrace{\pi_2}_{(A \wedge B) \rightarrow B} \left(\underbrace{K y}_{(A \wedge B) \rightarrow \neg B} \right) \right) : \neg(A \wedge B)$$

8. **Disjunction introduction** ($\Gamma, A \vdash A \vee B$ and $\Gamma, B \vdash A \vee B$). By the Deduction Theorem (Thm. 10) and axioms in_1/in_2 respectively.

9. **Negated disjunction introduction** ($\Gamma, \neg A, \neg B \vdash \neg(A \vee B)$). Recall that $\vdash \text{nege} : A \rightarrow \neg A \rightarrow B$, so by the Deduction Theorem (Thm. 10), we may construct $\vdash \text{nege}' : \neg A \rightarrow A \rightarrow B$ for arbitrary formulas A, B .

Now let C be any provable formula, e.g. $C = \top$ or $C = (z \rightarrow z)$, and let $\vdash P : C$. Then:

$$\Gamma, x : \neg A, y : \neg B \vdash \text{negi} \left(\underbrace{K P}_{(A \vee B) \rightarrow C} \right) \left(\underbrace{\text{match}(\text{nege}' x)}_{A \rightarrow C} \underbrace{(\text{nege}' y)}_{B \rightarrow C} \right) \quad \underbrace{\hspace{10em}}_{(A \vee B) \rightarrow \neg C}$$

10. **Double negation introduction** ($\Gamma, A \vdash \neg\neg A$).

$$\Gamma, x : A \vdash \text{negi} \left(\underbrace{K x}_{\neg A \rightarrow A} \right) \underbrace{\text{id}}_{\neg A \rightarrow \neg A} : \neg\neg A$$

11. **Proof by cases** (if $\Gamma, A \vdash B$ and $\Gamma, \neg A \vdash B$ then $\Gamma \vdash B$). By the Deduction Theorem (Thm. 10) and **Contrapositive** we have that $\Gamma \vdash P : \neg B \rightarrow \neg A$ and $\Gamma \vdash Q : \neg B \rightarrow \neg\neg A$.

$$\Gamma \vdash \text{dneg} \left(\underbrace{\text{negi } P Q}_{\neg\neg B} \right) : B$$

4 Soundness and completeness

Proposition 11 (Soundness). *If $\Gamma \vdash A$ then $\Gamma \models A$.*

Proof. By induction on the derivation of $\Gamma \vdash A$. There are three cases:

1. **Axiom.** It is routine to check that all the axioms are valid. For example, for the axiom K , let $V : \text{Var} \rightarrow \mathbf{2}$ be any valuation. Then $(A \rightarrow (B \rightarrow A))^V = (1 - A^V) \sqcup (1 - B)^V \sqcup A^V = 1$.
2. **Assumption.** Let $V : \text{Var} \rightarrow \mathbf{2}$ be any valuation such that $V \models \Gamma$. In particular $V \models A$ because $A \in \Gamma$ so we are done.
3. **Modus ponens.** The proof is of the form:

$$\frac{\Gamma \vdash B \rightarrow A \quad \Gamma \vdash B}{\Gamma \vdash A}$$

By *i.h.* we have that $\Gamma \models B \rightarrow A$ and $\Gamma \models B$. Let $V : \text{Var} \rightarrow \mathbf{2}$ be any valuation such that $V \models \Gamma$. Then $1 = (B \rightarrow A)^V = (1 - B^V) \sqcup A^V$ and $1 = B^V$, so we must have $A^V = 1$.

□

Lemma 12. *Let A be a formula depending (at most) on the first n propositional variables $\{p_1, \dots, p_n\}$. Let Γ be a context consisting of n formulas, the i -th of which is either p_i or $\neg p_i$. Then either $\Gamma \vdash A$ or $\Gamma \vdash \neg A$.*

Proof. By induction on A .

1. **Variable,** $A = p_i$. Then either $\Gamma \vdash p_i$ or $\Gamma \vdash \neg p_i$.

2. **Truth.** $\Gamma : \text{trivial} : \top$.
3. **Falsity.** $\Gamma : \text{negi} \underbrace{\text{abort}}_{\perp \rightarrow \top} \underbrace{\text{abort}}_{\perp \rightarrow \neg \top} : \neg \perp$.
4. **Negation,** $A = \neg B$. By *i.h.* there are two cases:
 - 4.1 If $\Gamma \vdash B$, by **Double negation introduction** $\Gamma \vdash \neg \neg B = \neg A$.
 - 4.2 If $\Gamma \vdash P : \neg B = A$ it is immediate.
5. **Conjunction,** $A = B \wedge C$. By *i.h.* on B there are two cases:
 - 5.1 If $\Gamma \vdash B$, then by *i.h.* on C there are two subcases:
 - 5.1.1 If $\Gamma \vdash C$, then by axiom pair we have $\Gamma \vdash B \wedge C = A$
 - 5.1.2 If $\Gamma \vdash \neg C$, then by **Negated conjunction introduction** $\Gamma \vdash \neg(B \wedge C) = \neg A$.
 - 5.2 If $\Gamma \vdash \neg B$, then by **Negated conjunction introduction** $\Gamma \vdash \neg(B \wedge C) = \neg A$.
6. **Disjunction,** $A = B \vee C$. By *i.h.* on B there are two cases:
 - 6.1 If $\Gamma \vdash B$, then by axiom in₁ we have $\Gamma \vdash B \wedge C = A$.
 - 6.2 If $\Gamma \vdash \neg B$, then by *i.h.* on C there are two subcases:
 - 6.2.1 If $\Gamma \vdash C$, then by axiom in₂ we have $\Gamma \vdash B \vee C = A$
 - 6.2.2 If $\Gamma \vdash \neg C$, then by **Negated disjunction introduction** $\Gamma \vdash \neg(B \vee C) = \neg A$.
7. **Implication,** $A = B \rightarrow C$. By *i.h.* on B there are two cases:
 - 7.1 If $\Gamma \vdash B$, then by *i.h.* on C there are two cases:
 - 7.1.1 If $\Gamma \vdash C$, then by **True consequent** we have $\Gamma \vdash B \rightarrow C = A$.
 - 7.1.2 If $\Gamma \vdash \neg C$, then by **Disproving an implication** we have $\Gamma \vdash \neg(B \rightarrow C) = \neg A$.
 - 7.2 If $\Gamma \vdash \neg B$, then by **False antecedent** we have $\Gamma \vdash B \rightarrow C = A$.

□

The previous lemma can be generalized when the context consists of $k \leq n$ formulas.

Lemma 13. *Let A be a formula depending (at most) on the first n propositional variables $\{p_1, \dots, p_n\}$. Let Γ be a context consisting of $k \leq n$ formulas, the i -th of which is either p_i or $\neg p_i$. Moreover, suppose that $\models A$. Then either $\Gamma \vdash A$ or $\Gamma \vdash \neg A$.*

Proof. By induction on k downwards from n to 0 (i.e. on $n - k$).

1. **Base case,** $k = n$. This is precisely the previous lemma (Lem. 12).
2. **Induction,** “ $k + 1 \implies k$ ”. Consider the context Γ extended with p_{k+1} . By *i.h.* there are two possibilities, $\Gamma, p_{k+1} \vdash A$ or $\Gamma, p_{k+1} \vdash \neg A$. The second case is not possible since, by Soundness (Prop. 11), we would have that $\Gamma, p_{k+1} \models \neg A$, i.e. there is a valuation V such that $V \models \Gamma, p_{k+1}$ and $A^V = 0$. But note that $\models A$ by hypothesis, so $A^V = 1$, a contradiction. Hence $\Gamma, p_{k+1} \vdash A$.

Similarly, if we consider the context Γ extended with $\neg p_{k+1}$ we obtain by *i.h.* that $\Gamma, \neg p_{k+1} \vdash A$.

Finally, note that $\Gamma, p_{k+1} \vdash A$ and $\Gamma, \neg p_{k+1} \vdash A$ together entail $\Gamma \vdash A$ using the principle of **Proof by cases**.

□

Theorem 14 (Completeness). *The following hold:*

1. **Completeness.** *If $\models A$ then $\vdash A$.*
2. **Implicational completeness.** *If $\Gamma \models A$ then $\Gamma \vdash A$.*

Proof. For item 1., by the previous lemma (Lem. 13) in the particular case in which $k = 0$, we have that either $\vdash A$ or $\vdash \neg A$. The second case is not possible since, by Soundness (Prop. 11), we would have that $\models \neg A$, *i.e.* that there is a valuation V such that $A^V = 0$. But note that $\models A$ by hypothesis, so $A^V = 1$, a contradiction. Hence $\vdash A$.

For item 2., suppose that $\Gamma \models A$. By the corollary of compactness (Coro. 6) there exists a finite subset $\{B_1, \dots, B_n\} \subseteq \Gamma$ such that $B_1, \dots, B_n \models A$. This in turn means that $\models B_1 \rightarrow \dots \rightarrow B_n \rightarrow A$. By item 1. of this theorem, we have $\vdash B_1 \rightarrow \dots \rightarrow B_n \rightarrow A$. By the Deduction Theorem (Thm. 10) we have $\Gamma' \vdash A$. Finally, by Weakening (Lem. 9), $\Gamma \vdash A$. □